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Modality-specific Noisy Label Learning for Multimodal Remote Sensing Image Classification

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***Abstract*—Multimodal remote sensing image classification has attracted significant attention due to its capability to integrate complementary information from diverse data sources. However, most existing methods assume that co-registered multimodal observations share a single clean semantic ground-truth label and do not consider the controlled setting in which each modality is associated with an independently corrupted supervision stream derived from that shared label. Existing noisy-label learning methods are primarily designed for a single supervision stream and are therefore not directly applicable to this setting. To address this issue, this paper proposes a robust multimodal remote sensing image classification framework tailored for scenarios involving modality-specific noisy label. The core of the proposed method consists of a class-adaptive dynamic sample selection strategy and a confidence-aware robust training strategy. Specifically, Gaussian Mixture Models are employed to estimate sample confidence for each modality individually, with class-adaptive dynamic thresholds determined through class-distribution adaptive modeling. Based on the confidence scores and thresholds from both modalities, training samples are dynamically divided into three separate groups. Each group undergoes a different multimodal joint training strategy to mitigate the impact of noisy labels. Furthermore, we incorporate inter-modal prediction consistency constraint and cross-modal contrastive learning to enhance semantic alignment and feature discriminability of the multimodal fusion networks. Experiments conducted on three public multimodal remote sensing image datasets (Houston2013, Berlin, Augsburg) under various simulated modality-specific noisy label conditions. The results demonstrate that the proposed method consistently outperforms state-of-the-art approaches across different noise types and intensities, exhibiting superior robustness and generalization.The code is available at https://github.com/AAAA-CS/MNLL.**

***Index Terms*—Remote sensing image classification, multimodal learning, noisy label learning, modality-specific noisy labels**

I. Introduction

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REMOTE sensing image classification is a key task in the intelligent extraction of geospatial information, with broad applications in land cover mapping, ecological monitoring, disaster warning, and urban planning [1, 2]. To improve classification accuracy, research efforts progressively shift from analyzing single-modal remote sensing imagery to integrating multimodal data, such as hyperspectral imagery (HSI), synthetic aperture radar (SAR), and light detection and ranging (LiDAR) [3, 4]. These modalities exhibit complementary characteristics in terms of imaging mechanisms, spatial resolution, and data representation, providing diverse and informative cues for constructing classification models with stronger discriminative power and generalization capability.

Although multimodal fusion presents significant potential for improving classification accuracy, it continues to face several practical challenges. One of the most critical challenges is the presence of label noise. Accurate annotation of remote sensing imagery requires expert knowledge and manual interpretation, which are not only costly and time-consuming but also susceptible to subjectivity and ambiguity in object boundaries. As a result, label errors are commonly observed [5]. In multimodal remote sensing scenarios, each modality is typically acquired using a distinct sensor and is subjected to different preprocessing and annotation workflows. These variations lead to inconsistencies in annotation quality across modalities [6]. For example, annotations in HSI may be compromised due to spectral mixing and similarity between material spectra, SAR labels may be biased by imaging distortions and speckle interference, and LiDAR annotations may be inaccurate due to sparse point clouds or occlusions. These practical factors define a distinctive learning scenario, namely multimodal learning with modal-specific noisy labels. In this scenario, noisy labels originate from heterogeneous sources and exhibit significant differences in both distribution and confidence levels across modalities.

Most current methods adopt a single fused label set as a unified supervision signal for multimodal data, often derived from a dominant modality or manually aligned across multiple modalities [7]. However, such an assumption of global label consistency neglects the differences and potential conflicts among modalities. This limitation hinders the capacity of models to address modality-specific noisy label. Moreover, most existing work on multimodal remote sensing classification focuses on designing for inter-modal feature interaction. Common strategies include feature-level fusion (early or mid-level) [8], decision-level fusion (late fusion) [9], and attention-based dynamic weighting mechanisms [10, 11]. While these

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approaches significantly improve the collaborative representation of multimodal features, they typically rely on the ideal assumption that all modalities share consistent and clean labels. In real-world applications, the inherent discrepancies in annotations across different modalities render this assumption invalid.

At the same time, numerous robust learning strategies have been developed in single-modal settings. These strategies aim to model and mitigate noisy labels originating from a single-modal data. Approaches include sample selection based on prediction consistency or loss distribution [12, 13], robust loss functions such as Generalized Cross Entropy (GCE) [14] and Normalized Cross Entropy (NCE) [15], and meta-learning methods that dynamically adjust model robustness [16]. Nevertheless, these methods generally assume that noisy label follows a homogeneous distribution, and they are not well-suited to handling the modality-specific and cross-modal propagation effects observed in multimodal scenarios. In practice, label errors are modality-specific, arising from the unique characteristics and processing pipelines of each sensor. Furthermore, within multimodal learning frameworks, these label errors may propagate through feature alignment or semantic fusion process, leading to cross-modal label error entanglement that further impedes learning. To date, there remains a lack of research explicitly addressing scenarios in which each modality possesses its own independently noisy label set. Existing methods have not yet fully leveraged inter-modal relationships to address this complex type of noisy label.

To address this challenge, we propose a robust modality-specific noisy label learning framework for multimodal remote sensing image classification, termed MNLL, specifically designed for scenarios in which each modality contains independently noisy labels. The framework consists of two key components: a class-adaptive dynamic sample selection and a confidence-aware robust training. Specifically, MNLL adopts a dual-branch fusion network architecture that outputs both fused predictions and modality-specific predictions. To mitigate the impact of noisy labels, a Gaussian Mixture Model (GMM) [17] is first applied to model the loss distribution of each modality independently, estimating the confidence of each sample being clean. A class-adaptive dynamic threshold estimation method is designed to account for class distribution disparities. Based on the sample confidence scores and class-specific thresholds from both modalities, the entire dataset is partitioned into three subsets; Each subset is assigned a tailored training strategy: confidence-weighted supervision for samples with high confidence in both modalities, robust loss optimization for those with high confidence in only one modality, and contrastive learning with augmented views for samples with low confidence in both modalities. Furthermore, to enhance the discriminative power and semantic consistency of the fused representations, a prediction consistency constraint and a cross-modal contrastive loss are introduced. The proposed method effectively exploits latent high-confidence consistent labels across modalities while leveraging low-confidence inconsistent samples to improve both robustness to label noise and generalization capability.

The main contributions of this work are summarized as follows.

1) This work investigates the complex scenario of multimodal learning where each modality contains independently noisy labels, explicitly challenging the conventional assumption of globally consistent annotations.
2) We propose a class-adaptive threshold modeling and dynamic sample selection strategy. By employing modality-specific GMM-based confidence estimation and class-adaptive dynamic thresholds, our method enables more effective use of modality-specific supervision signals.
3) We design a confidence-aware robust training strategy, which assigns tailored loss functions to three distinct sample subsets in order to fully exploit the available information from all samples, thereby enhancing both the reliability and discriminability of multimodal representations under modality-dependent noisy labels.
4) Extensive experiments on three publicly available multimodal remote sensing datasets demonstrate the superior performance of the proposed framework under various noise types and levels.

The remainder of this paper is organized as follows. Section II reviews related work on multimodal remote sensing image classification and learning with noisy labels. Section III presents the details of the proposed framework. Section IV provides experimental results and analysis based on three representative datasets. Section V concludes the study.

II. Related Works

A. Multimodal remote sensing image classification

With the continuous advancement of deep learning technologies, their applications in multimodal remote sensing image analysis have become increasingly widespread. These approaches have demonstrated significant advantages, particularly in high-level semantic understanding tasks such as land cover classification. Against this backdrop, existing methods for multimodal remote sensing image classification can be broadly categorized into two major groups: fusion strategy-based methods and feature interaction mechanism-based methods.

Fusion strategy-based methods are primarily distinguished by the stage at which multi-modal information is integrated within the neural network. Common strategies include early fusion and late fusion approaches [18]. Early fusion generally combines raw data or low-level features from different modalities at the input or shallow network layers, aiming to achieve low-level information integration. In contrast, late fusion extracts high-level features from each modality independently and then integrates the final prediction through classifier or decision-level fusion techniques. Representative works include the study by Ghamisi et al. [8], which first introduced deep learning-based fusion of HSI and LiDAR features, integrating multi-attribute extinction profiles with a convolutional neural network (CNN) classifier for effective feature representation. Chen et al. [19] employed multiple

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CNN branches to extract features from heterogeneous modalities and used a fully connected deep neural network for joint classification. Hang et al. [9] proposed a coupled dual-branch CNN framework, introducing fusion operations across several convolutional layers to integrate cross-modal features.

In contrast, feature interaction mechanism-based methods focus more on modeling the latent complementary relationships between modalities and guiding structural alignment. These methods typically design dedicated attention modules, feature interaction units, or alignment mechanisms to enable deep fusion and collaborative learning of heterogeneous modalities. For instance, Wu et al. [20] designed a cross-channel reconstruction (CCR) module embedded in the CNN backbone to enable explicit reconstruction and compact representation of inter-modal features. Ding et al. [21] proposed a Global-Local Transformer Network (GLT-Net), which combines the local modeling capabilities of CNNs with the long-range dependency modeling of Transformers to extract multi-scale features effectively. Zhang et al. [22] developed a Local Information Interaction Transformer (LIIT) structure, which jointly models the semantic coupling between HSI and LiDAR through a multi-branch embedding and local multi-source feature interaction module, and introduces a multi-source feature selection module for adaptive fusion. Wang et al. [23] designed a multi-attention hierarchical fusion network that integrates modality-specific attention and self-attention mechanisms to realize progressive feature fusion across shallow and deep layers. Roy et al. [10] introduced the Modality Fusion Transformer (MFT), which incorporates a multi-head cross-patch attention module and a cross-modal classification token to enable explicit information interaction and modality complementarity. Zhao et al. [24] proposed a multimodal Transformer framework that jointly models spectral-spatial features and modality-shared representations to enhance cross-source consistency. Chang et al. [25] developed the Dual Symmetric Fusion Transformer (DSymFuser), which incorporates local-global hybrid modules and symmetric fusion mechanisms at each layer to comprehensively capture collaborative relationships across heterogeneous modalities at multiple scales.

Aforementioned methods have achieved promising classification performance under clean label conditions. However, in real-world applications, label noise is inevitable, and the aforementioned models exhibit insufficient robustness to noisy labels during training.

B. Learning with noisy labels

To address the interference caused by noisy label, a large number of robust learning methods have been proposed in recent years [26]. These methods aim to enhance the learning ability and generalization performance of models in noisy environments. In general, existing approaches can be categorized into three main types. The first category focuses on the design of robust loss functions to alleviate the negative impact of noisy labels on model training. The second category introduces label correction mechanisms, which aim to improve the quality and effectiveness of training data by rectifying incorrect annotations. The third category is based on sample selection strategies, which identify and utilize more reliable training samples.

Natarajan et al. [27] systematically explored, from a theoretical perspective, how to correct arbitrary loss functions under class-conditional random label noise in order to improve their robustness. Subsequently, Ghosh et al. [28] pointed out that the mean absolute error (MAE) possesses inherent noise-tolerant properties under certain assumptions. However, MAE is difficult to optimize and tends to cause performance degradation in models. To strike a balance between robustness and optimization efficiency, Zhang et al. [14] proposed the GCE loss function, which combines the robustness of MAE with the implicit sample weighting mechanism of cross entropy, thereby achieving effective suppression of noise. Ma et al. [15] constructed the Active-Passive Learning (APL) framework, which dynamically enhances training stability by jointly optimizing two noise-robust loss functions. Xu et al. [29] proposed the Dual-Channel Residual Network (DCRN), where the loss function integrates an equal-weight combination of Reverse Cross Entropy (RCE) [30] and NCE [15] to improve robustness in noisy remote sensing image classification. Zhang et al. [31] introduced the Triple Contrastive Representation Learning (TCRL) framework, which integrates cluster-level, instance-level, and structure-level contrastive strategies to exploit feature dependency information for mitigating noise interference. Sarpong et al. [32] proposed the Dual Spectral-Spatial Residual Adaptive (DSSRA) network and designed the Adjustable Noise Tolerant Loss (ANTL) function. By combining normalized focal loss and RCE, this method effectively reduces the negative impact of noisy labels on model training. Although these loss-based robust learning strategies perform well in suppressing the overfitting of deep neural networks to noisy labels, they may still suffer from underfitting when dealing with clean samples or hard examples.

Some studies have approached the problem from the perspective of label correction, aiming to improve data validity and enhance model robustness. Tanaka et al. [33] utilized the average of historical predictions as correction targets. However, this approach suffers from unstable correction under high-noise conditions due to model overfitting. In contrast, Huang et al. [34] proposed a prototype-driven label correction framework, where noisy labels are aligned with prototype labels through class-balanced prototype queues. TCL [35] identified noisy samples as out-of-distribution samples using supervised GMM and explicitly corrected labels through cross supervision and entropy regularization, thereby achieving both label correction and feature alignment. Kim et al. [36] introduced the concept of structural labels and leveraged reverse k-nearest neighbors to mine local distributional structures for guiding feature manifold optimization and improving label quality. Shi et al. [37] proposed a hypergraph-driven label correction mechanism, where global-local hypergraph Laplacian energy minimization is employed to estimate label confidence and perform correction, enabling robust classification of remote sensing imagery. However, in complex multi-modal data scenarios, the presence of potential distribution shifts may undermine the effectiveness of these correction strategies,

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which still face certain generalization challenges.

In addition, some studies have focused on identifying cleanly labeled samples from noisy datasets. Existing work [38] has observed that deep neural networks tend to fit clean samples during the early stages of training and gradually overfit to noisy samples in later stages. As a result, clean samples generally exhibit lower loss in the early training phase. Based on this observation, Co-teaching and its variants [39, 40] train two peer networks using small-loss samples selected from each other, thereby mitigating the interference of noisy labels. DivideMix [41] was the first to introduce a GMM to model the loss distribution of training samples. This method partitions the dataset into labeled and unlabeled subsets and performs joint training using the MixMatch strategy [42] and a co-teaching mechanism. Zhang et al. [43] designed a training framework that integrates SPTNet with a noise-tolerant learning algorithm. By employing a confidence-driven sample partitioning strategy to distinguish clean and noisy samples, and incorporating a label-independent feature alignment regularization term, the framework effectively improves classification robustness for remote sensing imagery in the presence of noisy labels. DISC [44] proposed a dynamic instance-level selection strategy, which uses momentum-based thresholds derived from dual-view confidence and memory strength to dynamically classify samples into three subsets: clean, hard, and purified. Customized regularization strategies are then applied to each subset to enable graded utilization of noisy samples.

Most existing methods for learning with noisy labels are designed for single-modal noise scenarios. When directly applied to multimodal remote sensing tasks under modality-specific noise conditions, these methods encounter two critical limitations. First, they ignore the modality dependency of noisy label and often assume that noise is evenly distributed across modalities, which prevents accurate modeling of modality-specific noise characteristics. Second, they lack mechanisms for suppressing cross-modal error propagation, which can result in noise from one modality being transferred to others during feature alignment. This can cause semantic contamination and performance degradation. We propose a class-adaptive dynamic sample selection and a confidence-aware robust training method, which together effectively suppress modality-specific noisy label while promoting semantic alignment across modalities.

III. Proposed Methodology

The overall framework of the proposed MNLL is illustrated in Fig. 1. To address independently noisy labels across modalities in multimodal remote sensing images, we propose a robust classification framework comprising three core components.

1. Feature extraction. We extract feature representations for each modality using separate backbone networks $f(\cdot)$. The extracted features are then passed through the corresponding classification heads $F^1(\cdot)$, $F^2(\cdot)$, and the fusion classification head $F^{fus}(\cdot)$ to produce modality-specific predictions p^1, p^2, and the fused prediction p^{fus}, respectively. Meanwhile, the modality features are projected into low-dimensional embeddings z^1 and z^2 required for contrastive learning via the projection heads $H^1(\cdot)$ and $H^2(\cdot)$. The classification heads are implemented as single-layer multilayer perceptron (MLP), while the projection heads are designed as two-layer MLPs with batch normalization and ReLU activation functions. Both the classification and projection heads share the feature representations extracted by the respective modality backbones. (Section III-A)

2. Dynamic sample selection. To estimate label reliability, we model the loss distribution of each modality using GMM. Based on the estimated confidence and a class-adaptive dynamic thresholds, samples from each modality are dynamically divided into clean and noisy sets. By integrating confidence scores from both modalities, the training data is further partitioned into three subsets, which reflect varying degrees of label reliability across modalities. (Section III-B)

3. Robust training strategy. For each subset, we assign tailored training objectives. Dual-modality high-confidence samples use weighted supervision, partially confident samples introduce robust loss, and dual-modality low-confidence samples use contrastive reinforcement learning. Furthermore, by introducing prediction consistency constraints and cross-modal contrastive learning, we further enhance the discriminability of intra-modal features and improve the semantic alignment across modalities. (Section III-C)

Using hyperspectral data and LiDAR data as an example, we define the relevant notation as follows. Let $X^1 \in \mathbb{R}^{W\times H\times b^1}$ and $X^2 \in \mathbb{R}^{W\times H\times b^2}$ denote the hyperspectral and LiDAR modality data, respectively, where b^1 and b^2 represent the number of spectral bands for each modality. The symbols H and W denote the height and width of the image. At corresponding spatial locations in the two modalities, we select N image patches of size $l \times l$ as training samples. These are denoted as hyperspectral samples $x_i^1 \in \mathbb{R}^{s\times s\times b^1}, i \in \{1, 2, ..., N\}$ and LiDAR samples $x_i^2 \in \mathbb{R}^{s\times s\times b^2}, i \in \{1, 2, ..., N\}$. The corresponding ground-truth labels are given by $Y = \{y_1, y_2, ..., y_N\}$, where $y_i \in \{1, 2, ..., C\}$. To simulate a noisy label environment, we inject modality-specific noise into the two modalities to obtain noisy labels $\tilde{Y}^1 = \{\tilde{y}_1^1, \tilde{y}_2^1, ..., \tilde{y}_N^1\}$ for the hyperspectral modality and $\tilde{Y}^2 = \{\tilde{y}_1^2, \tilde{y}_2^2, ..., \tilde{y}_N^2\}$ for the LiDAR modality.

A. Feature extraction networks

We design a dual-branch multimodal remote sensing image classification network, as illustrated in Fig. 2. The overall architecture consists of three core components: a spectral-spatial joint feature extraction module, a multimodal feature extraction module, and a feature aggregation and classification module.

Spectral-spatial joint feature extraction module: First, a 1×1 convolution is applied to the hyperspectral input to perform nonlinear dimensionality reduction, reducing the original spectral channels from their full band count to 30 while preserving discriminative information and eliminating redundancy. This operation is followed by four layers of 3D convolution to extract local spectral structural features along

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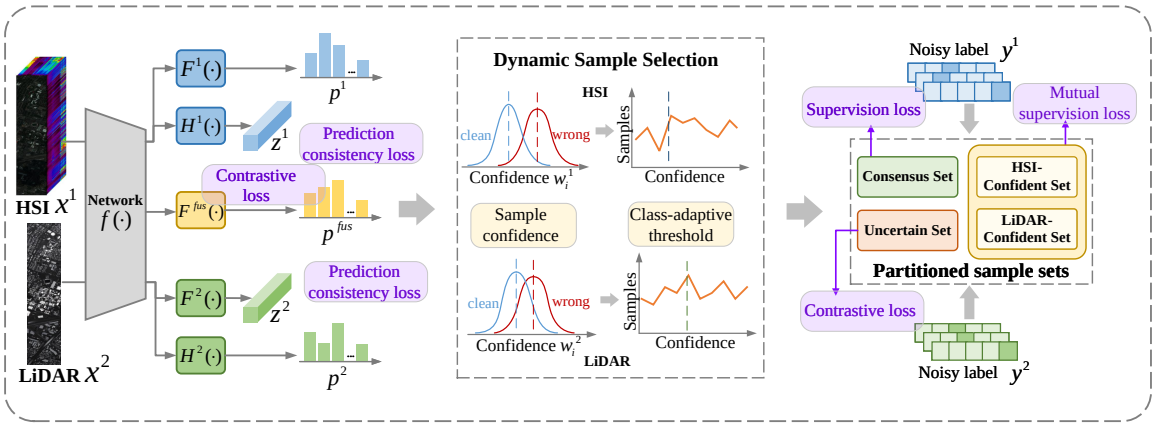


Fig. 1. The framework of MNLL. First, MNLL employs a dual-branch network to extract and fuse multimodal features. Then, a class-adaptive sample selection strategy, based on two-component GMM, is introduced to dynamically partition training samples into three subsets according to their estimated confidence values and class- adaptive thresholds. Finally, each subset is subsequently processed with tailored optimization strategies to effectively mitigate the impact of noisy label.

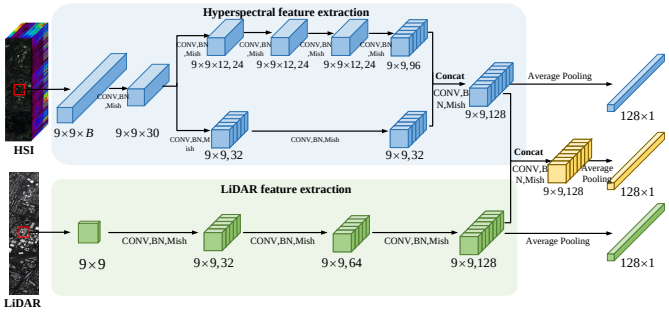


Fig. 2. Feature extraction networks.

the spectral dimension, and three layers of 2D convolution to capture spatial contextual information. The outputs of the two branches are concatenated along the channel dimension and further fused using a 1×1 convolution to produce the final spectral-spatial joint features. Multimodal feature extraction module: For the auxiliary modality (such as LiDAR), three layers of 2D convolution are used to progressively extract spatial structural information. These modality-specific features are then channel-wise fused with the hyperspectral features to form a comprehensive representation. Feature aggregation and classification module: The fused multimodal features are integrated using a 1×1 convolution and passed through global average pooling to obtain a unified feature representation. This representation is then input into a fully connected layer for classification. In addition, separate classifiers and projection heads are assigned to the hyperspectral and auxiliary modality branches.

B. Dynamic sample selection

In multimodal classification tasks, the presence of the noisy label and modality inconsistency label often leads to performance degradation. According to the findings in [39, 45], during the early stages of model training, the loss values of clean samples are consistently lower than those of noisy samples. Based on this observation, we propose a class-adaptive dynamic sample selection strategy, which jointly models the prediction loss of each sample across different modalities and dynamically selects reliable samples to enhance the robustness of model training.

First, we compute the cross-entropy loss for each sample in both modalities (HSI and LiDAR),

$$\ell_i{}^1 = L_{CE}(x_i^1, \tilde{y}_i^1), \ell_i{}^2 = L_{CE}(x_i^2, \tilde{y}_i^2) \tag{1}$$

To improve numerical stability, we first perform normalization on the loss vector of each modality $m \in \{1, 2\}$, respectively. For each modality, the normalized loss is fed into a corresponding two-component GMM to model the probabilistic distribution of the loss values. The GMM is composed of two Gaussian components, which are designed to capture the loss distribution of the underlying clean and noisy samples, respectively. The GMM parameters (i.e., mean, variance, and mixture weight) are iteratively estimated via the Expectation-Maximization (EM) algorithm, in which the E-step computes the posterior probability of each sample belonging to each component given the current parameters, and the M-step updates the parameters based on these posterior probabilities until convergence. Finally, the Gaussian component with the smaller mean is considered the latent clean distribution of samples, and the corresponding posterior probability $w_i{}^m$ of each sample on this component represents the confidence to which the sample belongs to the clean set.

Theoretically, the higher the confidence, the more likely the sample is to be clean. Therefore, based on confidence and thresholding setting, we can partition the samples from each modality into clean and noisy subsets. However, different classes exhibit varying learning difficulties and convergence rates, the fixed global threshold (applying identical thresholds across all classes) often leads to an imbalance where some classes select excessive samples as clean while others retain very few. To improve the reliability of sample partitioning, we propose a class-adaptive dynamic thresholding that adaptively predicts class-specific thresholds to achieve more accurate sample selection.

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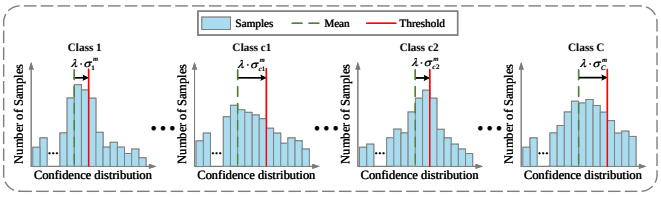


Fig. 3. Class-adaptive dynamic threshold. Each subplot depicts the clean confidence distribution of a class (blue histogram), with the green dashed line indicating the mean μ^m and the red solid line representing the dynamic threshold τ^m. Samples to the right of the threshold are considered clean, while those to the left are treated as noisy. The four representative cases—Low μ^m, Low τ^m; Low μ^m, High τ^m; High μ^m, Low τ^m; High μ^m, High τ^m—illustrate how the threshold adapts to varying distribution characteristics, from strict selection of only highly confident samples to more lenient retention under dispersed distributions. By tailoring the threshold to each class, this strategy avoids the misclassification risks inherent in global, uniform thresholds.

The prediction of class-adaptive dynamic threshold is shown in Fig. 3. For each original class c, we compute the mean and standard deviation of the confidences for all samples of that class, and use these statistics to define the confidence threshold for class c in the m-th modality as follows:

$$\tau_c^m = \mu_c^m + \lambda \cdot \sigma_c^m, \quad \lambda > 0 \tag{2}$$

where μ^m and τ^m represent the mean and standard deviation of the m-th modality in class c, respectively; λ is a hyperparameter that controls the strictness of the threshold. By incorporating the standard deviation into the mean, the intra-class confidence distribution can be adaptively reflected. We employ the class-specific mean as the baseline threshold and adjust it by incorporating the standard deviation. When the confidence distribution is dispersed, a relatively higher threshold ensures the reliability of selected samples; when the distribution is more concentrated, an excessively high threshold would exclude substantial clean samples, and therefore the threshold is maintained closer to the mean to improve the utilization of clean samples.

Based on the confidence (w_i^1, w_i^2) and the corresponding class-specific thresholds $(\tau_{\tilde{y}_i^1}^1, \tau_{\tilde{y}_i^2}^2)$ from both modalities, the training samples are partitioned into three subsets:

1) Dual-modality high-confidence samples: $w_i^1 > \tau_{\tilde{y}_i^1}^1$ and $w_i^2 > \tau_{\tilde{y}_i^2}^2$.

2) Single-modality high-confidence samples: ($w_i^1 > \tau_{\tilde{y}_i^1}^1$ and $w_i^2 < \tau_{\tilde{y}_i^2}^2$) or ($w_i^1 < \tau_{\tilde{y}_i^1}^1$ and $w_i^2 > \tau_{\tilde{y}_i^2}^2$).

3) Dual-modality low-confidence samples: $w_i^1 < \tau_{\tilde{y}_i^1}^1$ and $w_i^2 < \tau_{\tilde{y}_i^2}^2$.

During the subsequent training phase, we design different loss functions and supervision strategies for each subset. This enables the model to fully leverage the reliable samples while performing robust optimization on the unreliable ones.

C. Robust training strategy

In this section, we design a confidence-aware robust training strategy. Building upon the previously described dynamic sample selection method, the training samples are divided into three cases. We conduct different optimization designs for each individual case separately. Moreover, to enhance the discriminative capability and robustness of the multimodal fusion network, we further introduce a prediction consistency constraint and cross-modal contrastive learning applied to the entire set of samples. These components reinforce both semantic consistency and feature-level complementarity between modalities.

Dual-modality high-confidence samples. This type of sample exhibits high prediction confidence in both the HSI and LiDAR modalities and is therefore regarded as a strong supervision signal during training. However, even when a sample is confidently identified in both modalities, label inconsistency may still occur. To utilize such information robustly, we design a label consistency-aware supervision strategy. When the predicted labels from both modalities are consistent (i.e., $\tilde{y}_i^1 = \tilde{y}_i^2$), the sample is treated as a highly reliable "consensus sample," and standard cross-entropy supervision is directly applied to the fused prediction. The corresponding loss is defined as follows:

$$\mathcal{L}_i^{\text{agree}} = \mathcal{L}_{CE}\left(p_i^{fus}, \tilde{y}_i^m\right), \tag{3}$$

where, $p_i^{fus} = F^{fus}(f(x_i))$ denotes the prediction of the i-th sample in the fusion branch, and denotes an arbitrary modality. For samples with inconsistent labels (i.e., $\tilde{y}_i^1 \neq \tilde{y}_i^2$), we do not simply discard either label. Instead, we introduce a dual-label supervision strategy based on confidence weighting. Specifically, we first compute the normalized confidence from the two modalities as weighting coefficients $\alpha_i^1 = w_i^1/(w_i^1 + w_i^2), \alpha_i^2 = 1 - \alpha_i^1$. Based on these weights, we construct the following weighted cross-entropy loss:

$$\mathcal{L}_i^{\text{disagree}} = \alpha_i^1 \cdot \mathcal{L}_{CE}(p_i^{fus}, \tilde{y}_i^1) + \alpha_i^2 \cdot \mathcal{L}_{CE}(p_i^{fus}, \tilde{y}_i^2) \tag{4}$$

This strategy retains the supervisory information from both modality-specific labels while leveraging the confidence scores to guide the training process toward more reliable sources. When the confidence in one modality is significantly higher than in the other, its corresponding label receives a larger weight in the loss computation, thereby enabling adaptive focus on the more trustworthy supervision signal.

In summary, for dual-modality high-confidence samples, the final supervision loss is formulated as follows:

$$\mathcal{L}_i^{\text{clean}} = \begin{cases} \mathcal{L}_i^{agree}, & \text{if} \tilde{y}_i^1 = \tilde{y}_i^2 \\ \mathcal{L}_i^{disagree}, & \text{otherwise} \end{cases} \tag{5}$$

Single-modality high-confidence samples. When a sample exhibits high confidence in one modality while showing low confidence in the other, one modality considers the sample to be clean, whereas the other perceives it as potentially noisy. If both modality-specific labels are used to train the respective branches under such circumstances, it may easily lead to "incorrect learning" in the low-confidence modality. To address this issue, we treat the label from the high-confidence modality as a pseudo-supervisory signal and transfer it to guide the learning of the classification branch in the low-confidence modality. However, considering that the transferred label may

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still contain some noise, we do not employ the standard cross-entropy loss. Instead, we introduce a robust loss function composed of GCE [14] and NCE [15]:

$$\mathcal{L}_{\mathrm{GCE+NCE}}(p,y) = 1/2 * (\mathcal{L}_{\mathrm{GCE}}(p,y) + \mathcal{L}_{\mathrm{NCE}}(p,y)) \quad (6)$$

Among them, GCE is defined as follows:

$$\mathcal{L}_{\mathrm{GCE}} = \frac{1}{N}\sum_{i=1}^{N}\frac{1-(p_{i,y_i})^q}{q}, \quad q \in (0,1], \quad (7)$$

where, p_{i,y_i} denote the predicted probability of the i-th sample belonging to its ground-truth class y_i. Following [14], we also set $q = 0.7$. This loss function mitigates the impact of noisy labels by applying milder gradient penalties to low-probability samples, thereby improving the tolerance of model to noisy label.

NCE is defined as:

$$\mathcal{L}_{\mathrm{NCE}} = \frac{1}{N}\sum_{i=1}^{N}\frac{-\log p_{i,y_i}}{-\sum_{j=1}^{C}\log p_{i,j}} \quad (8)$$

NCE improves loss stability under class imbalance by normalizing the cross-entropy loss.

By applying the above robust loss to single-modality high-confidence samples, the loss for sample i-th is defined as follows:

$$\mathcal{L}_i^{rob} = \begin{cases} \mathcal{L}_{\mathrm{GCE+NCE}}\left(p_i^2, \tilde{y}_i^1\right), & \text{if} w_i^1 > \tau_{\tilde{y}_i^1}^1 \text{and} w_i^2 < \tau_{\tilde{y}_i^2}^2 \\ \mathcal{L}_{\mathrm{GCE+NCE}}\left(p_i^1, \tilde{y}_i^2\right), & \text{if} w_i^1 < \tau_{\tilde{y}_i^1}^1 \text{and} w_i^2 > \tau_{\tilde{y}_i^2}^2 \end{cases} \quad (9)$$

This strategy enables the model to maintain its guidance capacity for low-confidence modality outputs while effectively avoiding overfitting to noisy labels and achieving stable enhancement of feature representation of the low-confidence modality.

Dual-modality low-confidence samples. For this type of sample, both modalities exhibit low prediction confidence, making label quality unreliable and direct label supervision unjustified. Although the labels are untrustworthy, the samples still contain stable structural characteristics in their semantic representations, which can serve as effective signals for positive pairs in contrastive learning. Based on this, we generate two levels of augmented views for each modality. For instance, for the HSI modality, we apply random spatial rotation as weak augmentation, followed by random spectral shift to generate strong augmentation. For the LiDAR modality, we also apply spatial rotation as weak augmentation, while strong augmentation is achieved by injecting Gaussian noise to simulate sensor sampling errors. These four augmented views $\{x_i^{1,w}, x_i^{1,s}, x_i^{2,w}, x_i^{2,s}\}$ correspond to the weak and strong augmentations of the HSI and LiDAR modalities, respectively.

Each view is passed through the corresponding modality-specific projection head and mapped into the feature space for contrastive optimization. We employ the Normalized Temperature-scaled Cross Entropy (NT-Xent) loss based on cosine similarity to perform intra-modality contrastive learning. The loss function is defined as:

$$\mathcal{L}_{\mathrm{NT-Xent}}(z_i^w, z_i^s) = -\log\frac{\exp(sim(z_i^w, z_i^s)/\tau)}{\sum_{j=1}^{2N}\mathbf{1}_{[j\neq i]}\exp(sim(z_i^w, z_j)/\tau)}, \quad (10)$$

where, z_i^w and z_i^s respectively represent the feature representations of the same modality with weak and strong enhancement, $sim(\cdot,\cdot)$ is the cosine similarity, and τ is the temperature coefficient. The intra-modal contrastive loss for the i-th sample is defined as:

$$\mathcal{L}_i^{noisy} = 1/2*(\mathcal{L}_{\mathrm{NT-Xent}}(z_i^{1,w}, z_i^{1,s}) + \mathcal{L}_{\mathrm{NT-Xent}}(z_i^{2,w}, z_i^{2,s})) \quad (11)$$

This strategy effectively mitigates the risk of false supervision arising from label noise, thereby ensuring the reliability of training signals. Additionally, by incorporating a multi-view contrast enhancement strategy within each modality, the consistency and discriminative capability of modal-specific representation are significantly improved.

The differential training loss associated with the three dynamically selected sample subsets constitutes the primary supervision loss, as illustrated in equation 12.

$$\mathcal{L}^{\mathrm{sup}} = \mathcal{L}^{clean} + \mathcal{L}^{rob} + \mathcal{L}^{noisy} \quad (12)$$

The loss for each sample subset is averaged over its samples; if a subset is empty, its loss is set to zero, and the three subset losses use equal weights.

Prediction consistency constraint. To reduce the influence of inter-modal prediction bias on joint modeling, we implement prediction consistency constraints. This approach explicitly enhances the degree of semantic alignment between each modality branch and the fusion branch. The consistency loss is defined as follows:

$$\mathcal{L}^{cons} = \frac{1}{2N}\sum_{i=1}^{N}\left[\mathcal{L}_{\mathrm{BCE}}(p_i^1, p_i^{fus}) + \mathcal{L}_{\mathrm{BCE}}(p_i^2, p_i^{fus})\right] \quad (13)$$

This consistency loss is designed to strengthen the collaborative representation capability of the multimodal model. By minimizing the binary cross-entropy (BCE) between each modality output and the fusion output, it encourages the predictions from individual modalities to align more closely with the fusion result, thereby facilitating semantic collaborative guidance and enhancing the consistency of multimodal features. In scenarios involving noisy label or inconsistent modality predictions, this loss provides stable auxiliary supervision for the fusion branch, thereby mitigating the adverse effects of single modality errors on fusion performance.

Cross-modal contrast learning. We further introduce inter-modal contrastive learning to minimize the representation discrepancies between HSI and LiDAR data in the feature space. This constraint aims to draw the representation vectors of identical instances across different modalities closer together while effectively differentiating the representations of distinct sample instances. Its loss function is expressed as follows:

$$\mathcal{L}^{\boldsymbol{ctr}} = \frac{1}{N}\sum_{i=1}^{N} -\log\left(\frac{\exp\left((z_i^1)^T z_i^2/\tau\right)}{\sum_{j=1}^{N}\exp\left((z_i^1)^T z_j^2/\tau\right)}\right), \quad (14)$$

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where, τ is the temperature coefficient. To minimize the disparity in representation across modalities, this loss promotes a cohesive distribution of cross-modal representations for the same instance within the embedding space, while simultaneously distancing the representations of different samples.

The overall loss function is illustrated in equation 15.

$$\mathcal{L}^{total} = \mathcal{L}^{\sup} + \mathcal{L}^{cons} + \mathcal{L}^{ctr} \tag{15}$$

IV. Experiments

In this section, we conduct extensive experiments on three multimodal remote sensing datasets to evaluate the effectiveness of the proposed method for multimodal classification tasks with multi-source noisy labels. To comprehensively assess the performance, we adopt three widely used evaluation metrics: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. All experiments are repeated ten times with randomly selected training samples, and the mean and standard deviation of the results are reported.

A. Dataset Description

1) HoustonU2013 Data: The dataset covers the University of Houston campus and its surrounding regions in Texas, USA. The HSI, acquired by the ITRES CASI-1500 sensor, contains 349×1905 pixels with 144 bands. The LiDAR data provides elevation information over the same spatial extent as the HSI and contains only a single band. All the modalities have a spatial resolution of 2.5 meters per pixel. The dataset defines 15 distinct land-cover classes.

2) Berlin Data: The dataset covers the urban area of Berlin, Germany, and its surrounding rural regions. The original hyperspectral image has a spatial resolution of 797×220 pixels with 244 spectral bands ranging from 400 to 2500 nm. To match the spatial resolution of the SAR data, the HSI is resampled to 1723×476 pixels. The SAR data includes four feature channels. The ground-truth includes 8 distinct land-cover classes. Due to hardware memory limitations, the hyperspectral data is uniformly reduced to 60 dimensions using principal component analysis (PCA) in our experiments.

3) Augsburg Data: The dataset covers Augsburg, Germany and its surrounding regions. All imagery has been uniformly resampled to a 30-meter Ground Sampling Distance (GSD). The final image dimensions are 332×485 pixels, with the hyperspectral image containing 180 spectral bands covering the wavelength range of 0.4 μm to 2.5 μm. The SAR data includes four feature channels. The ground truth includes 7 classes. Table I shows the details of the three datasets.

B. Experimental Setting

All experiments are conducted on a hardware platform equipped with a GeForce RTX 2080 Ti GPU (11GB memory) using PyTorch 1.12.0 with CUDA 11.7 acceleration. The Adam optimizer is employed with an initial learning rate of 0.0006 and a batch size of 64. To ensure stable performance, all models are trained for 200 epochs.

Since each dataset contains only one ground-truth map shared by all modalities, we duplicate it and corrupt each copy with independent label noise to produce two observed supervision streams. Two types of label noise are considered: symmetric and asymmetric. Let the noise rate be denoted as ρ, representing the proportion of mislabeled samples in the training set. Under symmetric noise, samples from each class are randomly mislabeled as any other class with equal probability. Additional, we use a cyclic class-conditional transition matrix T (with $T_{ij} = p$ for $j = (i+1) mod C$) to generate asymmetric label noise.

TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE PU DATASET

HoustonU2013			Berlin			Augsburg		
Class	Class Name	Samples	Class	Class Name	Samples	Class	Class Name	Samples
1	Healthy Grass	1251	1	Forest	54954	1	Forest	13507
2	Stressed Grass	1254	2	Residential Area	268642	2	Residential Area	30329
3	Synthetic Grass	697	3	Industrial Area	19566	3	Industrial Area	3851
4	Tree	1244	4	Low Plants	59282	4	Low Plants	26857
5	Soil	1242	5	Soil	17426	5	Allotment	575
6	Water	325	6	Allotment	13305	6	Commercial Area	1645
7	Residential	1268	7	Commercial Area	24824	7	Water	1530
8	Commercial	1244	8	Water	6672			
9	Road	1252						
10	Highway	1227						
11	Railway	1235						
12	Parking Lotl	1233						
13	Parking Lot2	469						
14	Tennis Court	428						
15	Running Track	660						
Total		15029	Total		464671	Total		78294

C. Comparative experiments

To comprehensively evaluate the performance of this study, we conduct a comparative analysis with recently proposed noisy label handling methods in the field of remote sensing image classification, including DCRN [29], TCRL [31], SPTNet [43], and GLHLE [37]. In addition, we compare our method with state-of-the-art general-purpose noisy label learning algorithms such as TCL [35], LSL [36], and DISC [44]. Since these methods are originally designed for single-modality data, we employ the same multi-modal feature extraction network as our proposed method to ensure a fair comparison. We also evaluate the performance of two mainstream multi-modal remote sensing classification models, CCR [20] and MFT [10], under noisy label conditions. For all methods, we use the default parameters as specified in their respective publications. For each dataset, we conduct experiments under two different noise ratios. Considering that all baseline methods are designed for single-source label noise, we adopt the average classification accuracy under the two modality-specific noisy label settings as the evaluation metric for performance comparison.

1)Results on the HoustonU2013 Dataset: We randomly select 100 samples from each class to construct the training set and inject different types of label noise (symmetric or asymmetric) to simulate real-world data contamination. The remaining unselected samples serve as the test set. As shown in Table II, under the 30% symmetric noise setting, our approach consistently achieves the best performance. Table III shows the classification performance under two configurations with 30 % asymmetric noise. In multi-source noisy label scenarios, differences in label noise distributions across modalities are more likely to cause label inconsistencies, which in turn lead to the propagation and amplification of incorrect information during feature fusion. While our method achieves only a 3.66%

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TABLE II
AVERAGE CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE **HOUSTONU2013** DATASET WITH TWO SETS OF **30%** **MODALITY-SPECIFIC SYMMETRIC** NOISY LABELS. THE BEST RESULTS ARE MARKED IN BOLD.

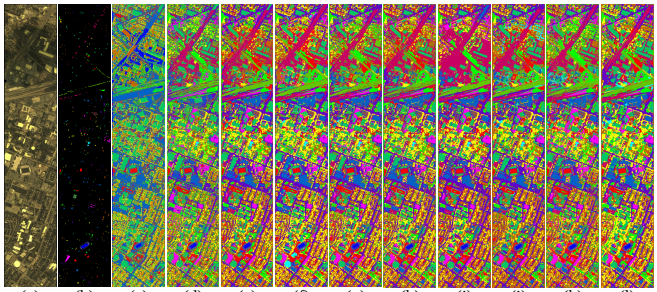
Class	CCR [20]	MFT [10]	DCRN [29]	TCRL [31]	SPTNet [43]	GLHLE [37]	TCL [35]	LSL [36]	DISC [44]	Ours
1	54.16±43.22	93.24±4.83	97.38±2.29	95.97±2.64	90.99±3.83	96.47±2.14	93.73±5.19	95.38±3.14	95.72±0.91	**97.48±1.51**
2	92.81±9.42	94.51±7.24	98.26±1.68	97.03±2.00	97.46±3.44	**99.39±0.04**	94.44±5.29	96.31±3.39	90.98±0.53	98.49±1.65
3	99.06±1.13	98.04±2.64	96.17±5.41	98.42±2.31	99.91±0.26	99.96±0.02	**99.99±0.01**	96.38±1.30	99.87±0.09	99.31±1.37
4	88.44±9.75	97.34±0.64	97.36±1.1	96.16±2.46	97.91±0.1	97.81±0.93	98.12±0.10	98.17±0.98	97.89±0.49	**98.18±0.62**
5	98.00±2.35	94.85±6.27	**99.84±0.12**	98.39±1.26	99.38±0.74	98.23±1.11	95.24±2.75	95.72±0.42	98.87±0.21	99.21±0.75
6	93.34±4.46	92.82±10.16	97.64±2.00	93.38±5.62	99.19±0.82	99.78±0.12	**99.86±0.24**	96.17±3.60	99.14±0.07	98.74±2.05
7	67.23±17.72	88.95±6.87	95.99±0.69	97.07±2.00	**98.56±1.62**	97.62±1.91	97.64±0.50	93.43±2.57	96.16±1.59	97.70±1.81
8	88.34±4.59	84.55±7.88	97.5±2.19	98.13±1.61	98.54±0.63	**99.38±0.50**	94.16±6.61	91.95±3.48	98.79±0.61	97.28±3.06
9	66.46±30.20	68.47±2.29	82.62±15.7	94.02±3.29	90.97±4.16	92.47±4.38	86.29±3.07	91.37±1.19	93.27±4.66	**95.90±2.08**
10	38.79±28.21	58.34±31.26	87.27±6.25	92.52±2.63	86.05±6.6	83.49±7.92	68.83±4.57	87.07±4.17	92.40±4.36	**93.79±2.69**
11	91.19±3.20	84.29±11.82	96.87±1.41	98.19±1.04	91.55±3.09	89.49±3.61	96.25±4.08	93.45±2.84	97.48±1.38	**98.24±0.54**
12	23.73±31.34	84.36±5.15	96.59±1.29	96.67±2.14	93.65±2.42	95.34±2.68	83.26±9.06	92.09±2.78	**96.99±1.73**	96.79±2.39
13	63.03±19.23	81.67±13.37	75.14±18.41	89.62±4.99	97.96±0.83	92.74±5.74	**98.87±1.29**	92.01±5.86	96.01±4.34	96.08±6.93
14	92.01±8.83	97.33±1.97	97.77±3.15	98.92±3.12	98.89±0.16	99.89±0.08	98.54±2.33	95.68±1.86	99.77±0.32	**99.93±0.14**
15	94.25±9.69	98.26±1.92	99.89±0.19	97.11±2.72	**99.89±0.08**	99.47±0.21	97.59±1.75	97.23±1.44	99.41±0.83	98.71±1.04
OA	73.85±3.93	87.7±1.89	93.82±2.85	96.19±0.67	95.06±1.90	95.44±1.83	91.42±1.28	94.13±1.54	96.29±2.24	**97.45±1.48**
AA	76.72±3.51	87.8±4.55	94.45±2.18	95.93±0.69	96.05±1.63	96.17±2.11	93.52±0.87	94.61±1.32	96.96±3.45	**97.72±1.61**
Kappa	71.75±4.30	86.68±2.05	93.31±3.09	95.88±0.72	94.65±1.98	95.06±1.98	90.71±1.39	93.72±1.59	95.98±2.63	**97.24±1.52**

TABLE III
AVERAGE CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE **HOUSTONU2013** DATASET UNDER **30%** **MODALITY-SPECIFIC ASYMMETRIC** AND **60%** **MODALITY-SPECIFIC SYMMETRIC** NOISY LABEL SETTINGS. THE BEST RESULTS ARE MARKED IN BOLD.

Noise Type	Assessment	CCR [20]	MFT [10]	DCRN [29]	TCRL [31]	SPTNet [43]	GLHLE [37]	TCL [35]	LSL [36]	DISC [44]	Ours
30% Asymmetric	OA	71.29±3.56	81.63±0.90	90.39±1.15	90.01±1.21	90.22±1.22	90.29±1.44	90.85±1.73	86.30±1.41	90.40±1.43	**93.79±2.33**
	AA	75.90±3.40	79.30±2.42	89.06±3.27	90.34±1.45	90.99±0.98	91.64±1.89	92.87±1.67	88.79±1.77	91.27±1.12	**94.19±2.20**
	Kappa	69.11±3.80	80.12±0.98	89.61±1.15	89.28±1.31	89.57±1.32	89.49±1.74	90.09±1.79	85.93±1.49	89.69±1.47	**93.28±2.52**
60% Symmetric	OA	49.69±2.99	78.07±1.67	81.97±2.07	85.19±2.21	88.34±2.63	90.03±2.14	82.24±2.51	83.52±3.48	90.16±1.30	**92.42±1.91**
	AA	51.69±3.73	77.96±1.66	80.62±4.45	84.75±2.08	90.03±2.09	91.19±2.47	86.00±1.52	87.26±3.19	91.81±1.51	**93.71±1.89**
	Kappa	45.88±2.88	76.27±1.83	80.52±2.21	83.99±2.39	87.36±2.67	90.37±1.92	80.76±2.53	82.16±4.94	89.35±1.32	**91.79±2.05**

drop in OA under asymmetric noise, we observe relatively higher variance across runs. This can be attributed to the systematic nature of asymmetric noise, where samples from certain classes are more likely to be mislabeled as specific confusing categories. Such non-uniform corruption introduces fluctuations in the effective training distribution across different runs, making class boundaries less stable and thereby increasing the standard deviation of performance.

To further evaluate model robustness under extreme noise conditions, Table III reports the results of all methods under two configurations with 60 % symmetric noise. The results show that general-purpose multi-modal remote sensing classification models such as CCR and MFT, which rely on cross-entropy loss, are highly sensitive to label noise and experience significant performance degradation. Although MFT improves feature representation through global information fusion and attention mechanisms and provides a certain degree of noise tolerance, its robustness under noisy conditions remains inferior to that of dedicated noisy label learning methods due to the lack of explicit noise-handling mechanisms. DCRN and TCRL enhance model robustness against noisy samples by designing different robust loss functions and perform well at a noise rate of approximately 30 %. However, under high noise conditions, relying solely on robust losses is insufficient to prevent the model from overfitting noisy labels. SPTNet leverages two networks to mine information from noisy samples and shows advantages in low-noise settings but fails to maintain performance under high noise levels. TCL detects noisy samples through supervised GMM modeling and



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 4. Classification maps for different methods on the HoustonU2013 dataset with two sets of 30% modality-specific symmetric noisy labels. (a) False-color map. (b) Ground Truth. (c) CCR. (d) MFT. (e) DCRN. (f) TCRL. (g) SPTNet. (h) GLHLE. (i) TCL. (j) LSL. (k) DISC. (l) Ours.

corrects labels via cross supervision and entropy regularization, making it suitable for moderate noise scenarios. LSL exploits local structural labels to guide feature learning and performs well under symmetric noise, but its generalization capability is limited when handling asymmetric or complex noise patterns. GLHLE constructs a global-local hypergraph structure to enhance label consistency modeling, and DISC dynamically partitions and utilizes samples based on dual-view confidence and memory mechanisms. Both methods exhibit strong robustness under single-modality noise conditions. The most methods suffer from significant performance degradation under high noise. Specifically, the OA of CCR drops by 24.16 %, MFT by 9.63 %, DCRN by 11.85 %, TCRL by 11.00 %, SPTNet by 6.72 %, GLHLE by 5.41 %, TCL by 9.18

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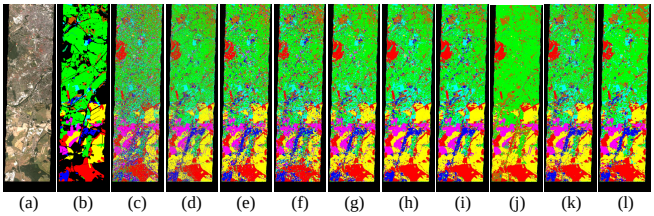


Fig. 5. Classification maps for different methods on the Berlin dataset with two sets of 30% modality-specific symmetric noisy labels. (a) False-color map. (b) Ground Truth. (c) CCR. (d) MFT. (e) DCRN. (f) TCRL. (g) SPTNet. (h) GLHLE. (i) TCL. (j) LSL. (k) DISC. (l) Ours.

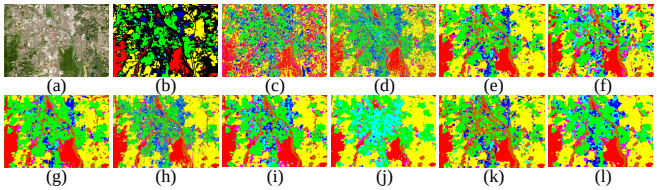


Fig. 6. Classification maps for different methods on the Augsburg dataset with two sets of 30% modality-specific symmetric noisy labels. (a) False-color map. (b) Ground Truth. (c) CCR. (d) MFT. (e) DCRN. (f) TCRL. (g) SPTNet. (h) GLHLE. (i) TCL. (j) LSL. (k) DISC. (l) Ours.

% , LSL by 10.61 % , and DISC by 6.13 % . In contrast, our method experiences only a 5.03 % drop in accuracy, which is significantly lower than all other compared methods.

This notable robustness and stability mainly result from our proposed modality-aware confidence modeling and collaborative optimization strategy. Specifically, we employ modality-independent GMM modeling and class-conditional dynamic thresholding to accurately estimate the confidence levels of samples in each modality. Based on this, we introduce a collaborative subset partitioning strategy to perform fine-grained grouping and differentiated training of multimodal samples, which effectively improves noise identification accuracy. Furthermore, the cross-modal collaborative supervision strategy fully utilizes high-confidence modality information to guide the optimization of low-confidence modalities, thereby suppressing the propagation and amplification of label noise across modalities. These components collectively enable our method to achieve stronger classification performance and more stable training under complex multi-source noisy label conditions.

Fig. 4 presents the complete classification maps of the HoustonU2013 dataset under two configurations with 30 % symmetric noise. It is observed that CCR and MFT tend to overfit noisy labels due to their reliance solely on the cross-entropy loss, resulting in a large number of misclassified regions and relatively low overall accuracy in the classification maps. In contrast, the classification results produced by MNLL appear smoother with clearer boundaries, indicating that it possesses strong noise resistance in complex multi-source noisy environments.

2) Results on the Berlin Dataset: We randomly select 300 samples from each class to construct the training set and inject different types of label noise to simulate real-world data contamination. Table IV reports the average results of all methods under two configurations with 30 % symmetric noise. For example, in terms of OA our method improves upon CCR by 36.11 % and outperforms MFT by 15.36 % , DCRN by 12.05 % , TCRL by 12.38 % , SPTNet by 9.07 % , GLHLE by 2.96 % , TCL by 3.39 % , LSL by 8.34 % , and DISC by 6.48 % . As shown in Table V, under asymmetric noise conditions, MNLL exhibits relatively minor performance degradation, which indicates its strong robustness and effectiveness in handling noisy labels. In addition, we also evaluate the robustness of MNLL under high-noise conditions. As reported in Table V, under two configurations with a noise ratio of 60 % , MNLL still maintains stable classification performance. Fig. 5 illustrates the complete classification maps generated by different methods on the Berlin dataset under two configurations with 30 % symmetric noise. Compared with other methods, the classification maps produced by the MNLL framework appear smoother and present clearer boundaries.

3) Results on the Augsburg Dataset: We randomly select 50 samples from each class to construct the training set. Table VI reports the average classification results of all methods under two configurations with 30 % symmetric noise. Taking OA as an example, our proposed method achieves an improvement of 42.29 % over CCR and outperforms MFT by 15.73 % , DCRN by 5.09 % , TCRL by 9.92 % , SPTNet by 0.30 % , GLHLE by 1.04 % , TCL by 3.65 % , LSL by 4.24 % , and DISC by 3.83 % . Table VII further presents the classification performance under asymmetric noise conditions. The MNLL demonstrates smaller performance degradation, highlighting its robustness and adaptability in handling non-uniform noisy labels. In addition, we evaluate the robustness of the method under higher noise levels. As shown in Table VII, when the noise ratio increases to 60 % , MNLL still maintains strong classification performance, with OA decreasing by only 13.53 % compared to the 30 % noise condition, which reflects stable noise resistance. Fig. 6 illustrates the complete classification maps on the Augsburg dataset under two configurations with 30 % symmetric noise. It can be observed that the map generated by MNLL appear smoother and exhibit clearer boundaries, further validating its superior performance in multi-source noisy environments.

D. Impacts of Free Parameters

1) The effect of different patch size: We evaluate the classification results under 30% symmetric noise across three datasets using different patch sizes ranging from 3×3 to 11×11, as shown in Fig. 7(a). The experiments reveal a clear unimodal trend: as the spatial window increases, the model accuracy initially improves and then declines, achieving optimal performance at a patch size of 9×9. A smaller patch size, such as 3×3, lacks sufficient contextual information and thus fails to effectively represent the spatial structure of complex land cover classes. In contrast, a larger patch size, such as 11×11, captures richer spatial dependencies but introduces excessive background noise, which significantly increases the risk of overfitting. Therefore, we select 9×9 as the optimal spatial scale.

2) The effect of temperature coefficient τ: The temperature coefficient τ is a critical hyperparameter in contrastive learning, which controls the distribution sharpness of sample

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TABLE IV
AVERAGE CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE **BERLIN** DATASET WITH TWO SETS OF **30%** **MODALITY-SPECIFIC SYMMETRIC** NOISY LABELS. THE BEST RESULTS ARE MARKED IN BOLD.

Class	CCR [20]	MFT [10]	DCRN [29]	TCRL [31]	SPTNet [43]	GLHLE [37]	TCL [35]	LSL [36]	DISC [44]	Ours
1	51.25±4.29	75.12±1.01	72.69±2.87	71.76±5.44	74.22±1.06	79.11±2.46	79.53±5.20	**85.63±10.19**	76.76±0.31	82.71±3.03
2	37.73±3.47	93.11±0.72	94.39±0.18	94.05±0.78	96.41±0.15	**96.52±0.47**	95.86±0.38	80.40±5.42	95.20±0.29	95.12±0.62
3	30.04±26.23	31.85±3.31	37.89±4.58	42.82±5.47	47.74±8.28	52.35±0.58	**53.38±5.92**	45.37±28.55	45.82±1.29	51.36±12.01
4	45.91±4.88	81.82±1.97	72.57±4.27	75.74±5.27	80.99±1.45	88.21±1.88	85.83±2.59	82.17±10.15	85.45±1.68	**90.77±2.09**
5	54.60±4.83	58.04±8.49	49.94±7.26	51.88±5.91	66.89±2.04	**81.13±0.25**	77.65±1.16	51.19±26.32	79.96±1.21	80.26±4.03
6	32.36±17.10	19.63±1.22	22.53±1.98	22.45±2.75	20.73±0.51	22.94±2.96	22.25±1.05	14.75±12.25	23.08±2.64	**32.52±4.71**
7	24.06±21.48	18.18±0.99	25.95±1.93	23.18±2.63	27.02±1.88	**32.92±3.31**	32.43±3.31	21.99±10.09	26.69±1.11	29.34±3.86
8	50.29±4.50	16.64±3.27	24.99±4.34	22.40±6.31	27.04±3.19	40.22±5.17	35.90±2.14	11.67±12.54	37.57±3.81	**61.88±14.32**
OA	39.98±1.99	60.73±1.39	64.04±1.86	63.71±2.37	67.02±1.44	73.13±1.53	72.7±1.35	67.75±4.17	69.61±1.99	**76.09±2.22**
AA	40.82±1.66	49.29±1.63	50.12±1.14	50.54±1.46	54.91±1.01	61.67±1.95	60.35±1.54	49.15±4.50	58.82±1.07	**65.43±3.61**
Kappa	24.87±1.26	48.52±1.32	52.16±1.95	51.42±2.18	55.63±1.47	63.24±1.61	62.47±1.37	48.53±5.57	58.99±2.04	**66.11±2.53**

TABLE V
AVERAGE CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE **BERLIN** DATASET UNDER **30%** **MODALITY-SPECIFIC ASYMMETRIC** AND **60%** **MODALITY-SPECIFIC SYMMETRIC** NOISY LABEL SETTINGS. THE BEST RESULTS ARE MARKED IN BOLD.

Noise Type	Assessment	CCR [20]	MFT [10]	DCRN [29]	TCRL [31]	SPTNet [43]	GLHLE [37]	TCL [35]	LSL [36]	DISC [44]	Ours
30% Asymmetric	OA	42.15±4.2	54.04±2.21	56.07±1.46	62.29±2.94	63.11±1.68	70.36±1.83	69.00±1.57	62.29±4.41	61.32±2.72	**74.19±3.83**
	AA	43.81±0.88	43.76±1.45	46.12±0.7	49.25±1.65	50.91±1.38	60.44±1.31	54.82±0.98	49.01±4.52	49.48±1.67	**63.33±2.94**
	Kappa	26.48±2.53	40.63±1.83	43.08±1.49	49.34±2.93	51.00±1.88	60.41±1.91	58.06±1.66	42.55±4.73	49.16±2.62	**63.75±4.25**
60% Symmetric	OA	25.58±8.56	37.05±4.06	34.66±3.29	36.06±1.95	39.54±2.25	55.98±2.17	47.40±3.05	49.34±5.49	40.08±2.59	**65.59±0.89**
	AA	21.38±2.26	31.94±2.79	28.22±1.87	29.08±0.77	32.43±1.04	49.87±3.41	39.71±1.05	42.52±3.59	33.77±1.18	**57.02±2.38**
	Kappa	8.11±4.58	23.65±3.77	20.86±3.13	21.91±1.37	26.09±1.82	45.36±2.85	35.44±2.28	36.67±4.33	27.82±2.08	**51.71±0.77**

TABLE VI
AVERAGE CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE **AUGSBURG** DATASET WITH TWO SETS OF **30%** **MODALITY-SPECIFIC SYMMETRIC** NOISY LABELS. THE BEST RESULTS ARE MARKED IN BOLD.

Class	CCR [20]	MFT [10]	DCRN [29]	TCRL [31]	SPTNet [43]	GLHLE [37]	TCL [35]	LSL [36]	DISC [44]	Ours
1	65.70±5.83	88.15±2.59	92.97±1.93	83.00±7.39	90.6±2.24	93.98±1.61	91.66±3.93	**97.51±0.69**	92.28±4.18	84.99±15.41
2	37.37±5.13	85.65±3.48	95.08±0.58	93.14±2.84	95.12±2.38	97.15±0.39	97.35±0.77	93.94±4.49	95.79±1.66	**97.79±0.47**
3	37.68±2.51	32.06±1.16	51.87±5.28	46.91±7.69	62.66±11.03	46.45±11.45	40.95±6.54	35.78±35.79	56.44±5.30	**65.20±3.94**
4	38.05±6.57	87.49±3.85	94.63±1.31	93.90±4.13	98.08±1.05	97.44±0.19	96.97±1.38	96.54±0.75	95.57±0.39	**99.37±3.10**
5	**55.73±6.89**	9.26±1.89	15.87±10.05	15.43±6.81	20.06±7.17	17.44±3.46	8.03±1.65	25.75±12.79	27.05±14.78	13.98±3.55
6	30.19±3.05	12.90±1.47	26.50±7.91	27.75±7.79	30.84±9.35	22.52±5.10	**43.59±14.93**	14.49±7.21	28.24±1.09	30.95±6.02
7	39.50±6.80	17.25±2.35	20.04±4.64	15.83±4.34	27.83±8.84	26.89±7.07	41.08±8.17	45.98±20.79	16.99±1.72	**63.06±27.75**
OA	42.53±2.04	69.09±1.93	79.73±4.19	74.90±2.21	84.52±1.86	83.78±2.75	81.17±2.25	80.58±3.37	80.99±2.31	**84.82±2.59**
AA	43.46±1.24	47.62±1.67	56.71±3.08	53.71±2.23	60.74±2.14	57.41±3.11	59.95±2.72	58.60±4.14	58.91±1.95	**65.05±3.27**
Kappa	29.67±1.97	58.96±2.29	72.76±5.18	66.58±2.61	78.73±2.33	77.94±2.97	74.62±2.81	73.57±4.40	74.32±2.78	**79.29±3.38**

TABLE VII
AVERAGE CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE **AUGSBURG** DATASET UNDER **30%** **MODALITY-SPECIFIC ASYMMETRIC** AND **60%** **MODALITY-SPECIFIC SYMMETRIC** NOISY LABEL SETTINGS. THE BEST RESULTS ARE MARKED IN BOLD.

Noise Type	Assessment	CCR [20]	MFT [10]	DCRN [29]	TCRL [31]	SPTNet [43]	GLHLE [37]	TCL [35]	LSL [36]	DISC [44]	Ours
30% Asymmetric	OA	42.57±3.9	55.16±3.09	68.3±3.58	71.97±2.29	78.89±2.58	80.03±2.03	78.97±2.63	76.55±3.16	75.01±3.21	**82.78±3.12**
	AA	42.69±2.86	43.19±4.72	51.39±2.6	52.91±1.98	58.05±3.05	55.46±2.44	58.36±3.36	51.78±1.44	55.12±2.45	**66.19±3.74**
	Kappa	28.48±4.54	43.49±4.84	59.13±4.48	63.00±2.88	71.83±3.16	73.39±2.71	71.85±3.33	67.59±3.79	67.18±3.88	**76.75±4.01**
60% Symmetric	OA	25.54±1.34	44.79±4.64	44.37±2.35	46.87±2.59	60.07±2.01	70.09±3.47	58.68±8.02	58.66±3.89	49.54±3.71	**71.29±3.56**
	AA	26.92±1.32	33.98±3.76	34.21±1.76	36.08±1.77	41.91±1.83	48.47±3.1	43.40±2.64	40.43±3.75	39.48±1.05	**51.96±2.01**
	Kappa	12.32±0.81	30.94±4.54	31.06±2.24	33.49±2.9	48.28±2.39	60.26±2.88	47.84±8.83	49.17±3.20	37.98±3.63	**62.63±3.89**

similarity. We set the value of τ in the range from 0.1 to 0.5 and conduct systematic comparative experiments on three datasets. The results are illustrated in Fig. 7(b). The experiments show that the model achieves the highest overall classification accuracy when $\tau = 0.25$. A smaller τ enhances the attraction between similar samples, making the model more sensitive to subtle feature differences. However, it may also lead to overfitting to locally similar samples during training. On the other hand, a larger τ reduces the distinction between positive and negative pairs, causing the similarity distribution to flatten and thus hindering the convergence efficiency of the contrastive objective. We finally adopt $\tau = 0.25$ as the default configuration.

3) The effect of parameter λ: The parameter λ determines the confidence threshold, which controls the strictness of clean sample selection. We adjust λ from 0.1 to 1.0 and evaluate the model on three datasets. As shown in Fig. 7(c), the model achieves the best classification performance when λ is set between 0.3 and 0.4. A smaller λ value corresponds to a more relaxed selection strategy, which may include low-confidence samples in the training process and increase the influence of noise. Conversely, a larger λ value causes the threshold to

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TABLE VIII
THE RESULTS OF ABLATION STUDIES ON VARIOUS PARTS OF LOSSES ON THREE DATASETS.

$\mathcal{L}^{clean}$	$\mathcal{L}^{rob}$	$\mathcal{L}^{noisy}$	$\mathcal{L}^{cons}$	$\mathcal{L}^{ctr}$	HoustonU2013			Berlin			Augsburg		
					OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
✓					87.4±3.86	89.06±3.95	86.28±3.07	68.92±3.11	59.95±4.79	57.96±3.01	81.75±3.54	60.71±3.72	74.95±4.76
✓	✓				94.41±1.34	94.47±1.29	93.95±1.97	73.04±3.86	61.66±4.53	60.75±3.84	82.49±3.75	61.45±3.19	78.17±4.81
✓	✓	✓			96.07±1.48	96.51±1.78	96.16±1.53	74.97±2.17	63.15±3.12	63.12±3.51	83.63±3.49	62.39±2.35	79.10±4.17
✓	✓	✓	✓		97.02±1.44	97.60±1.76	96.88±1.81	75.71±2.49	63.95±3.55	64.97±3.14	83.48±2.89	63.76±1.39	79.22±3.82
✓	✓	✓	✓	✓	**97.45±1.48**	**97.72±1.61**	**97.24±1.52**	**76.09±2.22**	**65.43±3.61**	**66.11±2.53**	**84.82±2.59**	**65.05±3.27**	**79.29±3.38**
Global threshold					95.65±1.66	96.19±1.60	95.29±1.71	71.07±4.01	61.97±2.80	60.03±4.92	82.56±3.12	62.77±3.99	76.03±4.03

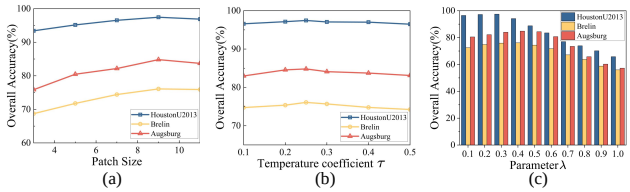


Fig. 7. The influence of free parameters on the overall accuracy of our method over the three datasets (30% modality-specific symmetric noisy labels). (a) Patch sizes. (b) Temperature coefficient τ. (c) Parameter λ.

rise too quickly, leading to the exclusion of many potentially useful samples. This significantly reduces the effective training set size and ultimately impairs the generalization ability of model. This trend is particularly evident on the HoustonU2013 dataset. Therefore, the choice of λ reflects the sensitivity of model to the density of the reliable sample distribution during training. Considering both sample utilization and model robustness, we recommend setting λ to 0.3 or 0.4.

E. Ablation Analysis

To systematically evaluate the individual contributions of our proposed modules to overall model performance, we conduct an ablation study on three datasets. All experiments are carried out under a 30% modality-specific symmetric noisy labels setting, and we report OA, AA, and the Kappa under different configurations, as shown in Table VIII.

First, in the baseline configuration where only the dual-modality high-confidence sample module is enabled, we utilize only samples that exhibit high confidence in both modalities and apply L^{clean} to the joint branch. Although this strategy effectively avoids noisy labels, the limited number of available samples leads to suboptimal performance across all datasets, indicating that relying solely on highly reliable supervisory signals is not sufficient to fully enhance model generalization. Subsequently, single-modal high-confidence Sample module is incorporated and a robust loss term L^{rob} is introduced, which significantly enhances model performance. For example, overall accuracy on the HoustonU2013 dataset increases by over 5%. These results indicate that the proposed cross-modal pseudo-supervision strategy together with the GCE-NCE joint loss effectively extract reliable supervisory signals from a single modality, thereby strengthening the discriminative capacity of the weaker modality. We then introduce the dual-modality low-confidence sample module, fully integrating samples identified via the confidence gating mechanism. By combining the weak and strong augmentation views within a contrastive learning framework, the model further improves on the Berlin and Augsburg datasets, achieving OAs of 74.97% and 83.63%, respectively. This result validates the effectiveness of the confidence-based dynamic sample selection strategy in guiding the model to extract discriminative features from potential hard samples. Building upon this, we add a prediction consistency constraint L^{cons} to further strengthen semantic alignment during multi-modal fusion. This mechanism explicitly encourages the joint branch to maintain consistency with modality-specific outputs in terms of predicted probabilities, thereby improving optimization stability in the early stages of training. On HoustonU2013, this mechanism yields additional performance gains, with OA reaching 97.02%. Finally, after integrating all modules and introducing the cross-modal contrastive loss L^{ctr}, the model achieves optimal performance on all three datasets, with OAs of 97.45%, 76.09%, and 84.82%, respectively. This module significantly reduces representational discrepancy between heterogeneous modalities. The improvement is particularly evident in Berlin, where modality differences are larger, underscoring the critical role of cross-modal semantic alignment in the fusion process. In summary, each module contributes substantially to enhancing both model robustness and discriminative power. In particular, the dynamic sample selection strategy combined with differentiated training strategies demonstrates superior generalization ability and training stability in noisy environments.

In addition, we conduct a comparison with a global threshold strategy, where a single threshold is computed for each modality based on the mean confidence of all samples, without considering class-specific variations. As shown in Table VIII, this approach yields inferior results compared to our class-conditional threshold. For example, on the Berlin dataset, the OA under the global threshold setting drops to 71.07%, which is 5.02% lower than our method with class-conditional thresholds. Similar trends are observed on HoustonU2013 and Augsburg. This performance gap highlights the importance of incorporating class-specific difficulty and convergence characteristics into the thresholding mechanism. By dynamically adapting thresholds for each class, our method achieves more precise sample partitioning, thereby preserving reliable supervision from hard-to-learn classes that would otherwise be misclassified under a global threshold.

F. Performance evaluation under different noise ratios

To evaluate the robustness of the proposed method under different levels of noise, we introduce symmetric label noise

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TABLE IX
PERFORMANCE RESULTS UNDER DIFFERENT NOISE RATIO COMBINATIONS ON THREE DATASETS.

			CCR [20]	MFT [10]	DCRN [29]	TCRL [31]	SPTNet [43]	GLHLE [37]	TCL [35]	LSL [36]	DISC [44]	Ours
HoustonU2013	0.1/0.3	OA	81.96±3.33	91.37±1.69	94.68±2.33	96.95±1.17	96.02±1.72	96.23±1.58	94.27±1.26	95.32±1.68	96.91±1.96	**97.51±0.64**
		AA	83.67±3.10	91.90±3.14	94.70±1.87	96.56±1.48	97.08±1.49	96.75±2.01	95.27±1.06	95.67±1.24	97.30±2.37	**97.81±0.68**
		Kappa	80.05±2.62	90.66±2.02	93.85±2.64	96.61±1.81	94.91±1.73	95.9±1.87	93.65±1.17	94.80±1.57	96.22±1.93	**97.34±0.71**
	0.3/0.6	OA	61.77±4.12	82.89±2.17	87.9±2.58	90.69±2.23	91.7±2.42	92.74±2.33	86.83±2.16	88.83±3.05	93.23±2.19	**95.59±1.68**
		AA	64.21±3.79	82.88±3.65	87.54±2.37	90.34±1.95	93.04±1.89	93.68±2.52	89.76±1.79	90.94±2.88	94.38±2.58	**96.23±1.83**
		Kappa	58.82±3.06	81.48±2.24	86.92±2.80	89.94±2.47	91.01±2.54	92.71±2.04	85.74±2.36	87.94±3.12	92.66±2.44	**95.22±2.74**
Berlin	0.1/0.3	OA	53.16±2.34	67.78±2.25	70.07±2.12	71.72±2.10	74.37±2.08	76.90±1.88	75.95±1.92	71.94±3.07	72.63±2.14	**77.81±2.76**
		AA	51.41±1.83	58.89±2.00	60.14±1.86	61.23±1.75	64.43±1.70	**67.98±1.63**	66.08±1.66	59.46±2.75	63.65±1.78	65.81±3.19
		Kappa	39.52±3.51	57.25±2.10	60.67±1.95	62.19±2.01	65.10±1.85	**69.59±1.73**	67.70±1.81	58.56±2.96	64.36±1.95	67.09±2.91
	0.3/0.6	OA	32.78±4.56	48.89±3.26	49.35±3.25	49.89±3.18	53.28±2.91	64.56±2.14	60.05±2.37	58.54±3.83	54.85±2.95	**68.75±2.95**
		AA	31.10±3.28	40.61±2.85	39.17±2.77	39.81±2.68	43.67±2.45	55.77±2.06	50.03±2.11	45.84±3.25	46.29±2.42	**59.61±3.83**
		Kappa	16.49±5.91	36.09±3.51	36.51±3.34	36.66±3.26	40.86±2.78	54.30±2.08	48.96±2.15	42.60±3.61	43.41±2.66	**56.22±2.94**
Augsburg	0.1/0.3	OA	62.10±1.81	76.32±1.75	82.35±3.84	82.21±2.51	87.58±2.45	86.41±3.10	84.48±2.87	73.47±3.48	84.55±2.21	**89.38±2.19**
		AA	49.93±1.17	53.21±1.49	59.83±2.78	60.14±2.62	62.48±2.17	59.38±3.66	62.64±3.13	60.27±3.67	63.23±2.03	**65.94±1.51**
		Kappa	51.88±1.84	68.01±2.27	75.60±3.95	74.20±2.84	83.16±2.53	83.36±3.41	78.86±3.44	79.53±3.82	79.63±2.82	**85.26±3.54**
	0.3/0.6	OA	34.04±1.93	56.94±3.85	62.05±3.56	60.89±2.83	72.30±2.35	76.94±3.62	69.93±4.54	69.62±4.16	65.27±3.49	**78.54±3.72**
		AA	35.19±1.77	40.80±3.27	45.46±2.39	44.90±2.26	51.33±1.97	52.94±3.59	51.68±3.72	49.52±4.03	49.20±1.87	**58.38±2.24**
		Kappa	20.99±1.26	44.95±3.96	51.91±3.31	50.04±2.88	63.51±2.21	69.10±3.96	61.23±4.82	61.37±5.19	56.15±3.17	**71.75±4.86**

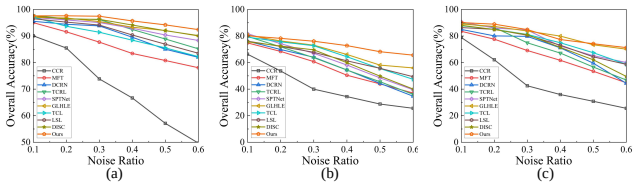


Fig. 8. OA value for the compared methods with noise ratios ranging from 0.1 to 0.6. (a) HoustonU2013 dataset; (b) Berlin dataset; (c) Augsburg dataset.

into the HoustonU2013, Berlin, and Augsburg remote sensing datasets, with noise ratios ranging from 0.1 to 0.6. As shown in Fig. 8, with the significant increase in noise levels, all comparison methods experience notable performance degradation. Typical multi-modal fusion methods, such as CCR and MFT, lack explicit noise-handling mechanisms. As a result, their OA drops sharply under high noise levels (e.g., $\rho = 0.6$), exhibiting severe overfitting. DCRN combines RCE and NCE losses and performs reasonably well under low noise, but its single robust loss becomes insufficient to maintain stable performance in high-noise environments. TCRL enhances feature consistency through multi-level contrastive learning, but its cluster-driven pseudo-supervision strategy lacks stability under severe noise interference. SPTNet selects reliable samples through confidence-based partitioning, but its modeling capacity is limited when facing complex noise distributions. The hyperedge construction process in GLHLE relies on feature accuracy and superpixel segmentation, which may lead to incorrect connections under high noise conditions. Furthermore, methods such as TCL, LSL, and DISC also struggle to maintain satisfactory performance in complex remote sensing scenarios with high levels of label noise. In contrast, the proposed method exhibits significantly stronger robustness and generalization across varying noise levels. The confidence-based dynamic sample partitioning effectively distinguishes between reliable and unreliable labels. The differentiated training strategy fully exploits the representational capacity of weakly supervised modalities. Additionally, the integration of cross-modal consistency and contrastive learning jointly guides semantic alignment across modalities, substantially alleviating interference caused by heterogeneous noise. On the HoustonU2013 dataset, as ρ increases from 0.1 to 0.6, the OA of our method decreases by only 5.47%. Similar robustness and stability are observed on the Berlin and Augsburg datasets, further verifying the effectiveness of the proposed method in highly noisy remote sensing scenarios.

G. Performance evaluation under different noise radio combinations

To further validate the robustness of the proposed method under heterogeneous noisy label conditions, this section conducts extended experiments on the HoustonU2013, Berlin, and Augsburg remote sensing datasets. We evaluate the model under two groups of modality-specific noise combinations (0.1/0.3 and 0.3/0.6). The results are presented in Tables IX.

Under the mild-to-moderate noise combination setting (0.1/0.3), all methods achieve relatively high performance. However, our method consistently yields the best results, demonstrating a clear advantage. Specifically, on the HoustonU2013 dataset, our method achieves 97.51% OA, 97.81% AA, and 97.34% Kappa, improving the OA by 2.83% and 0.56% compared to DCRN and TCRL, respectively, indicating strong robustness against cross-modal label disturbances. Similar trends are observed on the Berlin and Augsburg datasets, where our method leads in most metrics, confirming that the proposed class-conditional dynamic sample partitioning and differentiated training strategy effectively integrate multi-modal information and mitigate the adverse impact of low-quality label propagation. Notably, on the Berlin dataset under the 0.1/0.3 setting, our method achieves the highest OA, though AA and Kappa are slightly lower than GLHLE. This is because the dataset contains several semantically ambiguous classes with high inter-class similarity. Our sample selection

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strategy prioritizes high-confidence samples to avoid propagating noisy labels, which may under-represent difficult or boundary samples in these classes. As a result, AA and Kappa, which are sensitive to individual class performance, are slightly reduced, while overall scene-wide predictions remain robust. In the more challenging moderate-to-high noise combination setting (0.3/0.6), most existing methods exhibit significant performance degradation, while our method still maintains a notable advantage, demonstrating strong resistance to noise interference. For example, on the Berlin dataset, while many methods drop below 50% on the Kappa score, our method still achieves 59.61%, maintaining its lead under complex noise conditions.

In summary, our method consistently demonstrates superior robustness and generalization ability under both heterogeneous noise settings, verifying the effectiveness of its key components. On one hand, the confidence-based dynamic sample selection strategy distinguishes reliable supervision from noisy interference. On the other hand, the cross-modal consistency constraint and contrastive learning mechanisms effectively coordinate modality collaboration and enhance semantic alignment. This design enhances the capacity of model to maintain high-quality feature representation and generate robust decision outputs when confronted with challenges such as uneven label quality or multi-source errors, which are commonly encountered in practical remote sensing applications.

V. Conclusion

This study presents a robust multimodal remote sensing image classification framework designed to tackle the challenge of independently noisy labels in each modality. The proposed framework incorporates a class-adaptive dynamic sample selection strategy a confidence-aware training, enabling dynamic supervision tailored to modality-specific reliability. Training samples are divided into three categories based on confidence, with each category processed using tailored optimization strategies to mitigate the impact of noisy labels and enhance cross-modal consistency. Furthermore, by employing a dual-branch architecture along with consistency constraints and cross-modal contrastive learning, the method enhances feature discriminability and semantic alignment across modalities. Experiments conducted on three publicly available multimodal remote sensing image datasets under synthetically simulated conditions of independently noisy labels per modality. The results demonstrate that our framework significantly outperforms state-of-the-art methods.

While our framework demonstrates promising results under synthetic noise conditions, several limitations remain. The reliability of the current confidence estimation mechanism may be affected by the inherent instability of deep networks during the early training phase, potentially leading to inaccurate sample selection. Future work will focus on enhancing the robustness and efficiency of the confidence modeling process and validating the framework on datasets with real-world label noise to improve its practical applicability.

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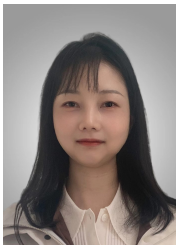
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